

Answer Key — Biology Grade 11

Sub-Strand 2.3: Excretion in Plants • Lesson 4: Cash from Waste: Categorising Plant Excretory Products and Their Value in Kenya

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	C	B	C	B	B	B

USE AFTER: OBSERVE PHASE — Part A: field observation (crushing a lemongrass or eucalyptus leaf to smell the essential oil)

Q1. During the field walk, a learner crushes a lemongrass leaf and smells a strong scent. Which type of plant excretory product is mainly responsible for this smell?

- A. Gums and resins
- B. Essential oils**
- C. Latex
- D. Tannins

Correct answer: B. Lemongrass and eucalyptus leaves hold essential oils in leaf glands, and crushing the leaf releases the scent. Tannins, latex and gums do not give this strong aromatic smell.

USE AFTER: OBSERVE PHASE — Part B: research cards and the table (Excretory product | Type | Plant source | Kenyan use)

Q2. Which row correctly matches the plant excretory product with its type and plant source?

- A. Gum arabic - alkaloid - Acacia senegal
- B. Papain - essential oil - papaya
- C. Rubber - latex - Hevea**
- D. Quinine - tannin - Cinchona

Correct answer: C. Rubber is made from latex tapped from the Hevea tree. Gum arabic is a gum (not an alkaloid), quinine is an alkaloid (not a tannin), and papain is a latex product from papaya (not an essential oil).

USE AFTER: EXPLAIN PHASE — Class master table and CER discussion (teacher corrects the 'latex is blood' misconception)

Q3. A learner says, 'Latex from a rubber tree is the blood of the plant.' Which statement correctly describes latex?

- A. It is a fluid that carries oxygen around the plant
- B. It is a milky fluid stored in special cells and vessels, and it is not blood**
- C. It is sugar solution made in the leaves and sent to the roots
- D. It is water and minerals moving up from the roots

Correct answer: B. Plants have no blood. Latex is a milky fluid stored in special latex cells or vessels, and it is an excretory product that people use to make rubber and papain.

USE AFTER: EXPLAIN PHASE — CER statement: 'Why are plant excretory products valuable in Kenyan markets?'

Q4. Complete the CER argument. Claim: Plant excretory products are valuable in Kenyan markets. Evidence: quinine from Cinchona bark, rubber from Hevea latex and gum arabic from Acacia. Which is the best reasoning?

- A. The products are valuable because they are the main food that plants make for growth
- B. The products are valuable because they come only from plants that grow in Kenya
- C. Plants store these products in tissues such as bark and latex vessels, and people use their chemical properties, such as treating malaria, sealing and sticking**
- D. The products are valuable because plants throw them away, so they are free and easy to collect

Correct answer: C. Good reasoning links the accumulation of wastes in plant tissues to the useful chemical properties humans exploit, such as antimalarial action, stickiness and sealing. The other options give wrong ideas about what the products are or why they are valuable.

USE AFTER: DQB PHASE — Revisiting the Driving Question Board (moving a question to 'answered' with evidence)

Q5. A group wants to move the DQB question 'Why is the sticky sap from the Acacia tree worth money?' to 'answered'. Which evidence from today best justifies the move?

- A. The sap oozes out only when the tree is cut, so it has no use
- B. Acacia sap hardens into gum arabic, which collectors in Turkana and Marsabit sell, so a plant waste earns money**
- C. All liquids that come out of plants are sold in markets
- D. Acacia trees have thorns and grow in dry areas of Kenya

Correct answer: B. The sap hardens into gum arabic, an excretory product that people collect and sell. This links the phenomenon directly to its economic value. The other options are unrelated or incorrect.

USE AFTER: MODEL BUILDING PHASE — Revising the class model: labelled arrows showing where excretory products end up, plus 'Value to people'

Q6. A group is updating its class model of plant excretion. Which annotation is correct?

- A. Latex is made in the leaf glands
- B. Gums and resins ooze out of stem wounds**
- C. Essential oils are stored in latex vessels in the stem
- D. Tannins are found only in the root hairs

Correct answer: B. Gums and resins ooze from wounds in the stem, as seen with Acacia. Essential oils are held in leaf glands, latex is held in latex vessels and bark, and tannins are found in bark and leaves.

USE AFTER: END OF LESSON — Daily Exit Ticket: 'Contrast guttation and transpiration in 2 sentences'

Q7. Which statement correctly contrasts guttation and transpiration in plants?

- A. Both are the loss of latex from cut stems, but they happen at different times of day
- B. Guttation is loss of liquid water droplets from leaf edges (often on humid mornings), while transpiration is loss of water vapour, mainly through stomata**
- C. Guttation is loss of water vapour through stomata, while transpiration is loss of liquid droplets from leaf edges
- D. Guttation happens only in old yellow leaves, while transpiration happens only in young leaves

Correct answer: B. Guttation releases liquid droplets along leaf edges, as seen on grass and arrowroot on humid mornings. Transpiration is the loss of water vapour, mainly through the stomata.